

Questions marked with (*) should be skipped by multichance students who have requested a 30% reduction in the exam length.

Exercise 1

Consider the problem

$$\begin{cases} -\operatorname{div}(\alpha \nabla u) + \gamma u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (1)$$

where $\alpha > 0$ and $\gamma \geq 0$ are two given constant coefficients, while f is a given function and $\Omega \subset \mathbb{R}^2$.

- 1.1. **[1 pt]** Write the weak formulation of problem (1).
- 1.2. **[2 pt]** Prove existence and uniqueness of the solution of the problem obtained at point 1.1.
- 1.3. **[2 pt]** Approximate (1) by the finite element method, using polynomials of degree 2 and provide the corresponding stability and error estimate.
- 1.4. **[2 pt]** (*) Partition the domain Ω into two disjoint subdomains Ω_1 and Ω_2 and write one iteration of the Dirichlet–Neumann method. Is this method convergent?
- 1.5. **[2 pt]** Using `deal.II`, implement a finite element solver for problem (1) that uses the Dirichlet–Neumann method with relaxation, with the following settings:

$$\Omega = (0, 3) \times (0, 1), \quad \Omega_1 = (0, 1) \times (0, 1), \quad \Omega_2 = (1, 3) \times (0, 1),$$

$$\alpha = 1, \quad \gamma = 1, \quad f(x) = 1.$$

The meshes for the two subdomains can be downloaded at

<https://github.com/michelebuccelli/nmpde-labs-aa-25-26/tree/main/lab-08/mesh>.

Use linear finite elements. Upload the source code of the solver.

- 1.6. **[2 pt]** (*) Set the maximum number of Dirichlet–Neumann iterations to 15. For the two relaxation values $\lambda = 1.0$ and $\lambda = 0.25$, upload a plot of the solution at the final iteration and a plot of the solution along the line $y = 0.5$.

Exercise 2

Consider the problem

$$\left\{ \begin{array}{ll} \frac{\partial \mathbf{u}}{\partial t} - \mu \Delta \mathbf{u} + \nabla p = 0 & \text{in } \Omega, \ t \in (0, T), \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, \ t \in (0, T), \\ \mathbf{u} = 0 & \text{on } \Gamma_D, \ t \in (0, T), \\ \mu \frac{\partial \mathbf{u}}{\partial n} - p \mathbf{n} = \psi & \text{on } \Gamma_N, \ t \in (0, T), \\ \mathbf{u} = 0 & \text{in } \Omega, \ t = 0, \end{array} \right. \quad (2)$$

where $\mu > 0$, ψ is a function depending on both $\mathbf{x}$ and t , $\Omega \subset \mathbb{R}^2$, $\Gamma_D \cap \Gamma_N = \emptyset$, $\Gamma_D \cup \Gamma_N = \partial\Omega$, and $T > 0$.

- 2.1. **[3 pt]** (*) Write the weak formulation of (2).
- 2.2. **[1 pt]** Write the spatial approximation of (2) using the Taylor–Hood finite element spaces of degree 3 for the velocity $\mathbf{u}$ and 2 for the pressure p .
- 2.3. **[2 pt]** Approximate the problem obtained at point 2.2 by the implicit backward Euler method in time. Then explain why the problem obtained at every timestep has a unique solution.
- 2.4. **[2 pt]** Write the expected error estimate for the problem obtained in point 2.3 in terms of both Δt and h .
- 2.5. **[4 pt]** Using `deal.II`, implement a finite element solver for problem (2). Use Taylor–Hood finite elements of degree 2 for the velocity and 1 for the pressure. Use the mesh found at <https://github.com/michelebuccelli/nmpde-labs-aa-24-25/blob/main/examples/gmsh/mesh-pipe.msh> and depicted in Figure 1.

Consider the following data:

$$\Gamma_D = \Gamma_2 \cup \Gamma_3, \quad \Gamma_N = \Gamma_0 \cup \Gamma_1,$$

$$\psi(x, t) = -p_0(x)\mathbf{n},$$

where

$$p_0(x) = \begin{cases} 1, & \text{on } \Gamma_0, \\ 2, & \text{on } \Gamma_1. \end{cases}$$

and

$$\mu = 1, \quad T = 1.$$

Upload the source code of the solver.

- 2.6. **[3 pt]** Compute the solution to the problem, and upload:
 - (a) a plot of the solution at the final time T ;

(b) a plot of the velocity magnitude profile along

$$R = \Omega \cap \{y = x\},$$

that is along the line $y = x$ (Hint: use the ParaView filter “Plot over line”);

(3.) (*) a plot of the velocity magnitude over time at the point

$$\mathbf{x} = (1.25, -0.4)^T$$

(Hint: combine the ParaView filters “Probe location” and “Plot data over time”).

Figure 1: Domain for Exercise 1. Each boundary Γ_i has tag i in the mesh file.